

Unit 1 Guided Notes — The Real Number System

Classify it, convert it, estimate it, and order it. Every number has a home in the real number system.

Standards: NC.8.NS.1 · NC.8.NS.2

This packet has three parts: **Part 1 — Guided Notes** (fill in the blanks as you learn), **Part 2 — Practice Problems**, and **Part 3 — Answer Key** (do not peek until you've finished!). Work the blanks on your printed copy first, then click any blank, step, or check button on screen to reveal the answer and check your thinking.

Topics Covered: 1.1 Converting Numbers & Repeating Decimals (NC.8.NS.1) · 1.2 Number Classification (NC.8.NS.1) · 1.3 Perfect Squares, Cubes & Their Roots (NC.8.NS.2) · 1.4 Converting Repeating Decimals to Fractions (NC.8.NS.1) · 1.5 Estimating Square & Cube Roots (NC.8.NS.2) · 1.6 Ordering Rational & Irrational Numbers (NC.8.NS.2)

1 Converting Numbers & Repeating Decimals

Fraction → Decimal: DIVIDE the numerator by the denominator.

Fill in each decimal on your printed copy, then click to check.

Fraction	Division	Decimal
$\frac{3}{4}$	$3 \div 4$	
$\frac{1}{8}$	$1 \div 8$	
$\frac{1}{3}$	$1 \div 3$	

✦ **REPEATING DECIMALS:** When division keeps going with the same pattern, use a BAR over the repeating digits.

💡 **KEY INSIGHT:** Repeating decimals are because they can be written as fractions!
Non-repeating, non-terminating decimals (like π) are .

Decimal → Fraction Conversions

Every terminating decimal can be written as a fraction — proof that it is **rational**.

Decimal	How	Fraction (simplest form)
0.75	Read it, write it, simplify	$\frac{75}{100} = \frac{\quad}{\quad}$
0.6	Read it, write it, simplify	$\frac{6}{10} = \frac{\quad}{\quad}$
0.125	Read it, write it, simplify	$\frac{125}{1000} = \frac{\quad}{\quad}$

Quick Check

MC Which decimal is equivalent to the fraction $\frac{3}{8}$?

✓ Divide: $3 \div 8 = 0.375$, a terminating (rational) decimal.

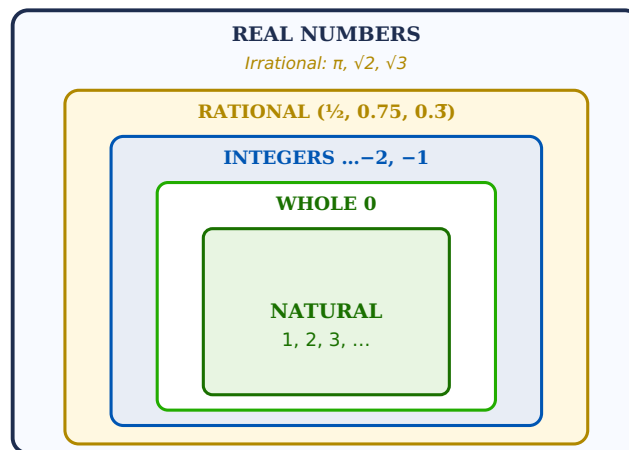
2 Number Classification

Every real number belongs to one or more of these sets. Fill in the definition for each:

Set	Examples	Definition
Natural Numbers	1, 2, 3, 4, ...	numbers

Whole Numbers	0, 1, 2, 3, ...	Naturals +
Integers	..., -2, -1, 0, 1, 2, ...	Whole numbers +
Rational Numbers	$\frac{1}{2}$, 0.75, $-\frac{3}{5}$, 0.3	Can be written as a (includes terminating and repeating decimals)
Irrational Numbers	π , $\sqrt{2}$, $\sqrt{3}$	CANNOT be a fraction; , decimal

Real Number System — Nested Set Diagram



Match the Set to its Definition

MATCH Click a set on the left, then its definition on the right.

Set

Natural Numbers

Whole Numbers

Integers

Irrational Numbers

Definition

Whole numbers + their negatives

Counting numbers: 1, 2, 3, ...

Non-repeating, non-terminating decimal

Naturals + zero

0 of 4 matched

Quick Check

MC Which number is irrational?

✓ 7 is not a perfect square, so $\sqrt{7}$ is a non-repeating, non-terminating decimal — irrational.

ERROR ANALYSIS A student claims:

" $\sqrt{49}$ is irrational because it has a square root symbol." **What's the mistake?**

The square-root symbol does **not** make a number irrational. $\sqrt{49} = 7$, a whole number — so it is **rational**. A root is irrational only when the number under it is *not* a perfect square (like $\sqrt{7}$ or $\sqrt{2}$).

3 Perfect Squares, Cubes & Their Roots

✦ **THE RELATIONSHIP:** Squaring and square roots are INVERSE OPERATIONS — they undo each other! If $5^2 = 25$, then $\sqrt{25} = 5$. Know one, know the other!

Memorize 1-15

Fill the squares on your printed copy, then click to check.

n	n^2	n	n^2	n	n^2
3		4		5	
8		9		10	
13		14		15	

✦ **CUBES & CUBE ROOTS:** Cubing and cube roots are also INVERSE OPERATIONS. If $5^3 = 125$, then $\sqrt[3]{125} = 5$. The cube-root symbol carries a small **3**: $\sqrt[3]{}$.

Memorize the Perfect Cubes 1-10

Fill the cubes on your printed copy, then click to check.

n	n^3	n	n^3
1		6	
2		7	
3		8	
4		9	
5		10	

🔗 **SQUARE ROOT vs CUBE ROOT:** $\sqrt{64}$ asks "what number times itself = 64?" → 8. $\sqrt[3]{64}$ asks

“what number used three times = 64?” $\rightarrow 4$. Same radicand, different answer — watch the little 3!

Worked Example — Click each step to reveal

4-Step Algorithm: Estimate $\sqrt{20}$

Step 1 – Find perfect squares: $16 < 20 < 25 \rightarrow \sqrt{16} = 4, \sqrt{25} = 5$

Step 2 – Numerator & denominator: $20 - 16 = 4$ (numerator); $25 - 16 = 9$ (denominator) $\rightarrow$ fraction = $4/9$

Step 3 – Divide: $4 \div 9 \approx 0.44$

✓ **Step 4 – Add:** $4 + 0.44 = 4.44 \rightarrow \sqrt{20} \approx 4.4$

$$\sqrt{n} \approx \text{lower root} + (n - \text{lower}^2) \div (\text{upper}^2 - \text{lower}^2)$$

4 Converting Repeating Decimals to Fractions — The 9's Rule

✦ **THE 9's RULE:** The repeating digits go in the NUMERATOR. The DENOMINATOR uses 9's — one 9 for each repeating digit!

Repeating Digits	Denominator	Example	Simplified
1 digit		$0.\overline{3} = 3/9$	$= \frac{\quad}{\quad}$
2 digits		$0.\overline{27} = 27/99$	$= \frac{\quad}{\quad}$
3 digits		$0.\overline{125} = 125/999$	$= \frac{\quad}{\quad}$

Worked Example — Convert 0.27 (0.272727...) to a fraction

Step 1 — Count repeating digits: 2 digits repeat (the digits 27)

Step 2 — Write denominator with that many 9's: Denominator: 99

Step 3 — Numerator = repeating block: 27/99

✓ Step 4 — Simplify: $27/99 = 3/11$

⚠ **IMPORTANT:** The 9's rule only works when the decimal STARTS repeating right after the decimal point. For decimals like 0.1666... (where 6 repeats but 1 doesn't), use the algebraic method. ALWAYS simplify your final fraction!

5 Estimating Square & Cube Roots

✦ **THE BIG IDEA:** Most square roots are irrational — they don't come out evenly. We can estimate them using two methods!

Method 1: Number Line Placement

Step 1: Find the two consecutive **perfect squares** your number falls between.

Step 2: Take the **square root** of each — those are your two consecutive integers.

Step 3: Decide: Is your number closer to the **lower** or **upper** perfect square?

Step 4: Place it on the number line **accordingly**.

Method 2: The 4-Step Algorithm

$$\sqrt{n} \approx \text{lower root} + (n - \text{lower}^2) \div (\text{upper}^2 - \text{lower}^2)$$

Practice: Estimate $\sqrt{45}$

Step 1: $36 < 45 < 49 \rightarrow \sqrt{36} = 6, \sqrt{49} = 7$

Step 2 – Fraction: $(45 - 36) \div (49 - 36) = 9 \div 13$

Step 3: $9 \div 13 \approx 0.69$

✓ **Step 4:** $6 + 0.69 = 6.69 \rightarrow \sqrt{45} \approx 6.7$ (Actual: 6.708)

🔗 WHICH METHOD TO USE?

- **Number Line:** Great for seeing roughly WHERE the root falls — perfect for "between which integers" questions.
- **Algorithm:** Great for getting a decimal approximation — more precise for "estimate to the nearest tenth."
- Both start with the same Step 1: find the two perfect squares that surround your number!

Your Turn — Calculation

CALC Estimate $\sqrt{30}$ to the nearest tenth using the 4-step algorithm.

$\approx ?$

Estimating Cube Roots

✦ **SAME IDEA, PERFECT CUBES:** To estimate a cube root, find the two perfect cubes it falls between, then estimate to the nearest tenth.

Worked Example — Estimate $\sqrt[3]{30}$

Step 1 – Perfect cubes: $27 < 30 < 64 \rightarrow \sqrt[3]{27} = 3, \sqrt[3]{64} = 4$

Step 2 – Which is it closer to? 30 is just above 27, so the answer is just above 3

✓ **Step 3 – Estimate:** $\sqrt[3]{30} \approx 3.1$ (check: $3.1^3 \approx 29.8$ ✓)

CALC Estimate $\sqrt[3]{70}$ to the nearest tenth. (Hint: $64 < 70 < 125$)

$\approx ?$

Negative Square Roots

✦ **NEGATIVE ROOTS:** A negative sign in front simply flips the estimate to the other side of zero. Estimate the positive root first, then make it negative. Example: $-\sqrt{40} \rightarrow \sqrt{40} \approx 6.3$, so $-\sqrt{40} \approx -6.3$.

CALC Estimate $-\sqrt{55}$ to the nearest tenth.

$\approx ?$

6 Ordering Rational & Irrational Numbers

✦ **THE STRATEGY:** To compare and order a mix of fractions, decimals, mixed numbers, and irrational numbers — convert everything to decimals first!

Worked Example — Order: $\sqrt{5}$, $2\frac{1}{4}$, $\frac{7}{3}$, 2.1

Number	Decimal Approximation	How?
$\sqrt{5}$	$\approx$ _____	Use 4-step algorithm
$2\frac{1}{4}$	$=$ _____	$2 + (1 \div 4) = 2.25$
$\frac{7}{3}$	$\approx$ _____	$7 \div 3 = 2.333\dots$
2.1	$=$ _____	Already a decimal

Order from least to greatest: _____ < _____ < _____ < _____



✦ **MIXED NUMBERS:** Convert to improper fractions OR decimals. Decimals are usually faster!
 Mixed number $\rightarrow$ Decimal: whole number + (numerator $\div$ denominator). Example: $3\frac{2}{5} = 3 + (2 \div 5) = 3 + 0.4 = 3.4$

⚠ WATCH OUT:

- When comparing, make sure you're using the same number of decimal places — it's easy to mis-read 3.375 vs 3.5 if you look at them quickly!
- Irrational numbers like $\sqrt{7}$ need to be estimated first before you can place them in order.

ERROR ANALYSIS To order 1.5, $\sqrt{2}$, $\frac{7}{4}$, $\sqrt{3}$, a student wrote $1.5 < \sqrt{2} < \sqrt{3} < \frac{7}{4}$. Are they right? Explain.

Convert first: $1.5 = 1.5$, $\sqrt{2} \approx 1.41$, $\frac{7}{4} = 1.75$, $\sqrt{3} \approx 1.73$. The student forgot that $\sqrt{2} \approx 1.41$ is the *smallest*. Correct order: $\sqrt{2} < 1.5 < \sqrt{3} < \frac{7}{4}$.

🔑 PART 2 — Practice Problems

Show all work on your printed copy. Circle your final answer, then check on screen.

A Classify Each Number

Remember: Rational = can be written as a/b (fractions, terminating decimals, repeating decimals). Irrational = non-repeating, non-terminating decimals.

1 $0.333\dots$

✓ It equals $1/3$; a repeating decimal is always rational.

2 π

✓ $\pi = 3.14159\dots$ is non-repeating and non-terminating.

3 $\sqrt{49}$

✓ $\sqrt{49} = 7$, a perfect-square root is a whole number.

4 $\sqrt{10}$

✓ 10 is not a perfect square, so $\sqrt{10}$ is irrational.

5 $-3/8$

✓ Any fraction of integers is rational: $-3/8 = -0.375$.

6 $0.1212\dots$

✓ The digits 12 repeat, so this equals $12/99 = 4/33$.

7 $\sqrt{144}$

✓ $\sqrt{144} = 12$, a perfect-square root.

8 $0.101001\dots$

✓ The pattern grows but never repeats — irrational.

9 $2/7$

✓ $2/7 = 0.285714\dots$, a repeating decimal (fraction = rational).

10 $\sqrt{2}$

✓ 2 is not a perfect square. $\sqrt{2} \approx 1.41421\dots$

11 $-\sqrt{81}$

✓ $-\sqrt{81} = -9$, a whole number.

12 3.14

✓ 3.14 is a terminating decimal = $314/100$. Rational!

B Estimate Each Square or Cube Root (Nearest Tenth)

How: For square roots, find the perfect squares above & below; for cube roots, use perfect cubes. Estimate to the nearest tenth. A leading – just flips the sign.

1 $\sqrt{5}$

≈ ?

2 $\sqrt{10}$

≈ ?

3 $\sqrt{20}$

≈ ?

4 $\sqrt{30}$

≈ ?

5 $\sqrt{50}$

≈ ?

6 $\sqrt{8}$

≈ ?

$\approx ?$

7 $\sqrt{15}$

$\approx ?$

8 $\sqrt{40}$

$\approx ?$

9 $\sqrt{65}$

$\approx ?$

10 $\sqrt{90}$

$\approx ?$

11 $\sqrt[3]{30}$

$\approx ?$

12 $\sqrt[3]{100}$

$\approx ?$

13 $-\sqrt{50}$

$\approx ?$

C Order from Least to Greatest

Strategy: Convert each number to a decimal approximation, then arrange from least to greatest. Show your decimal conversions!

1. Order: π , $\sqrt{10}$, $10/3$, 3.5

Decimals: 3.14, 3.16, 3.33, 3.5 $\rightarrow \pi < \sqrt{10} < 10/3 < 3.5$

2. Order: 1.5, $\sqrt{2}$, $7/4$, $\sqrt{3}$

Decimals: 1.5, 1.41, 1.75, 1.73 $\rightarrow \sqrt{2} < 1.5 < \sqrt{3} < 7/4$

3. Order: $\sqrt{5}$, 2.5, $11/5$, $\sqrt{7}$

Decimals: 2.24, 2.5, 2.2, 2.65 $\rightarrow 11/5 < \sqrt{5} < 2.5 < \sqrt{7}$

4. Order: 4.2, $\sqrt{18}$, $13/3$, $\sqrt{20}$

Decimals: 4.2, 4.24, 4.33, 4.47 $\rightarrow 4.2 < \sqrt{18} < 13/3 < \sqrt{20}$

5. Order: -2.5, $-\sqrt{5}$, $-8/3$

Decimals: -2.5, -2.24, -2.67 $\rightarrow -8/3 < -2.5 < -\sqrt{5}$

6. Order: $\sqrt{50}$, 7.1, $22/3$, $\sqrt{52}$

Decimals: 7.07, 7.1, 7.33, 7.21 $\rightarrow \sqrt{50} < 7.1 < \sqrt{52} < 22/3$

7. Order: $5/6$, $\sqrt{0.5}$, 0.9, $7/8$

Decimals: 0.833, 0.707, 0.9, 0.875 $\rightarrow \sqrt{0.5} < 5/6 < 7/8 < 0.9$

8. Order: $\sqrt{30}$, 5.5, $17/3$, $\sqrt{32}$

Decimals: 5.48, 5.5, 5.67, 5.66 $\rightarrow \sqrt{30} < 5.5 < \sqrt{32} < 17/3$

★ PART 3 — Answer Key ★

Do not distribute to students until after the lesson.

★ Guided Notes — Answers

1.1 Converting Numbers

$3/4 = 0.75$ · $1/8 = 0.125$ · $1/3 = 0.333... = 0.\overline{3}$ · Repeating decimals are **RATIONAL**; non-repeating, non-terminating decimals are **IRRATIONAL**. $0.45 \rightarrow 45\%$ · $75\% \rightarrow 0.75$ · $0.75 = 75/100 = 3/4$.

1.2 Number Classification

Natural = Counting numbers · Whole = Naturals + zero · Integers = Whole numbers + negatives · Rational = can be written as a fraction · Irrational = non-repeating, non-terminating.

1.3 Perfect Squares & Cubes

$3^2=9$, $4^2=16$, $5^2=25$, $8^2=64$, $9^2=81$, $10^2=100$, $13^2=169$, $14^2=196$, $15^2=225$. Cubes: $1^3=1$, $2^3=8$, $3^3=27$, $4^3=64$, $5^3=125$, $6^3=216$, $7^3=343$, $8^3=512$, $9^3=729$, $10^3=1000$. $\sqrt[3]{125} = 5$ (inverse of cubing). $\sqrt{20}$: $16 < 20 < 25 \rightarrow 4 + 4/9 \approx 4.44 \rightarrow \approx 4.4$.

1.4 The 9's Rule

1 digit $\rightarrow /9$ ($0.\overline{3} = 1/3$) · 2 digits $\rightarrow /99$ ($0.\overline{27} = 27/99 = 3/11$) · 3 digits $\rightarrow /999$ ($0.\overline{125} = 125/999$).

1.5 Estimating Square & Cube Roots

Square roots — find the two perfect squares around it, then estimate. $\sqrt{45}$: $36 < 45 < 49 \rightarrow 6 + 9/13 \approx 6.69 \rightarrow \approx 6.7$. $\sqrt{30} \approx 5.5$. Cube roots — use perfect cubes: $\sqrt[3]{30}$ ($27 < 30 < 64$) ≈ 3.1 ; $\sqrt[3]{70}$ ($64 < 70 < 125$) ≈ 4.1 . Negative roots flip the sign: $-\sqrt{55} \approx -7.4$.

1.6 Ordering

$\sqrt{5} \approx 2.24$, $2\frac{1}{4} = 2.25$, $7/3 \approx 2.33$, $2.1 = 2.1 \rightarrow 2.1 < \sqrt{5} < 2\frac{1}{4} < 7/3$.

Part A — Classify

1 Rational ($=1/3$) · 2 Irrational · 3 Rational ($\sqrt{49}=7$) · 4 Irrational · 5 Rational (-0.375) · 6 Rational ($=4/33$) · 7 Rational ($\sqrt{144}=12$) · 8 Irrational · 9 Rational · 10 Irrational · 11 Rational (-9) · 12 Rational ($=314/100$).

Part B — Estimate (nearest tenth)

$\sqrt{5} \approx 2.2$ · $\sqrt{10} \approx 3.2$ · $\sqrt{20} \approx 4.5$ · $\sqrt{30} \approx 5.5$ · $\sqrt{50} \approx 7.1$ · $\sqrt{8} \approx 2.8$ · $\sqrt{15} \approx 3.9$ · $\sqrt{40} \approx 6.3$ · $\sqrt{65} \approx 8.1$ · $\sqrt{90} \approx 9.5$ · $\sqrt[3]{30} \approx 3.1$ · $\sqrt[3]{100} \approx 4.6$ · $-\sqrt{50} \approx -7.1$.

Part C — Order Least to Greatest

1) $\pi < \sqrt{10} < 10/3 < 3.5$ 2) $\sqrt{2} < 1.5 < \sqrt{3} < 7/4$ 3) $11/5 < \sqrt{5} < 2.5 < \sqrt{7}$ 4) $4.2 < \sqrt{18} < 13/3 < \sqrt{20}$ 5) $-8/3 < -2.5 < -\sqrt{5}$ 6) $\sqrt{50} < 7.1 < \sqrt{52} < 22/3$ 7) $\sqrt{0.5} < 5/6 < 7/8 < 0.9$ 8) $\sqrt{30} < 5.5 < \sqrt{32} < 17/3$.

← Back to Unit 1